Manipulating time data

2026. February 27., Friday 14:42

Adding and subtracting date / time data

```
SELECT date '2005-09-11' - date '2005-09-10';
```

```
+---------+
| integer |
|---------|
| 1       |
+---------+
```

Subtracting date values is always an integer

Adding and subtracting date / time data

```
SELECT date '2005-09-11' + integer '3';
```

```
+------------+
| date       |
|------------|
| 2005-09-14 |
+------------+
```

Adding integers to date values the implied precision is days

Adding and subtracting date / time data

```
SELECT date '2005-09-11 00:00:00' - date '2005-09-09 12:00:00';
```

```
+----------------+
| interval       |
|----------------|
| 1 day 12:00:00 |
+----------------+
```

For time, we get an interval

AGE()

Calculating time periods with AGE

```
SELECT AGE(timestamp '2005-09-11 00:00:00', timestamp '2005-09-09 12:00:00');
```

```
+----------------+
| interval       |
|----------------|
| 1 day 12:00:00 |
+----------------+
```

Date / time arithmetic using INTERVALs

```
SELECT timestamp '2019-05-01' + 21 * INTERVAL '1 day';
```

```
+----------------------------+
| timestamp without timezone |
|----------------------------|
| 2019-05-22 00:00:00        |
+----------------------------+
```

If we want to do it with timestamps we can do INTERVALs, which we can multiply/divide too

<u>Current date and time</u>

NOW()-timestamp with microsecond precision **with timezone**
SELECT NOW()::timestamp returns it without timezone
Or we can do SELECT CAST(NOW() as timestamp)

Or CURRENT_TIMESTAMP, where we can specify precision with CURRENT_TIMESTAMP(2)

If we do not require timestamp precision:
SELECT CURRENT_DATE
SELECT CURRENT_TIME

Manipulating timestamp data
Useful when we want to use only year, month, and we may not care about precision

Extracting and transforming date / time data

- EXTRACT(**field** *FROM* **source**)

```
SELECT EXTRACT(quarter FROM timestamp '2005-01-24 05:12:00') AS quarter;
```

- DATE_PART('**field**', **source**)

```
SELECT DATE_PART('quarter', timestamp '2005-01-24 05:12:00') AS quarter;
```

Day of the week = dow
Source needs to be a valid timestamp, time or interval data type
EXTRACT() and DATE_PART() can be used interchangeibly

Extracting sub-fields from timestamp data

Data from *payment* table by year and quarter

```
SELECT
  EXTRACT(quarter FROM payment_date) AS quarter,
  EXTRACT(year FROM payment_date) AS year,
  SUM(amount) AS total_payments
FROM
  payment
GROUP BY 1, 2;
```

Results

```
+---------------------------------+
| quarter | year | total_payments |
|---------|------|----------------|
| 2       | 2005 | 14456.31       |
| 3       | 2005 | 52446.02       |
| 1       | 2006 | 514.18         |
+---------------------------------+
```

Watch GROUP BY here: it groups by first and second column - ORDER BY can do the same

Truncating timestamps using DATE_TRUNC()

The `DATE_TRUNC()` function will truncate timestamp or interval data types.

- Truncate timestamp '2005-05-21 15:30:30' by year

```
SELECT DATE_TRUNC('year', TIMESTAMP '2005-05-21 15:30:30');
```

```
Result: 2005-01-01 00:00:00
```

- Truncate timestamp '2005-05-21 15:30:30' by month

```
SELECT DATE_TRUNC('month', TIMESTAMP '2005-05-21 15:30:30');
```

```
Result: 2005-05-01 00:00:00
```

Returns an interval or timestap rather than a numerical value